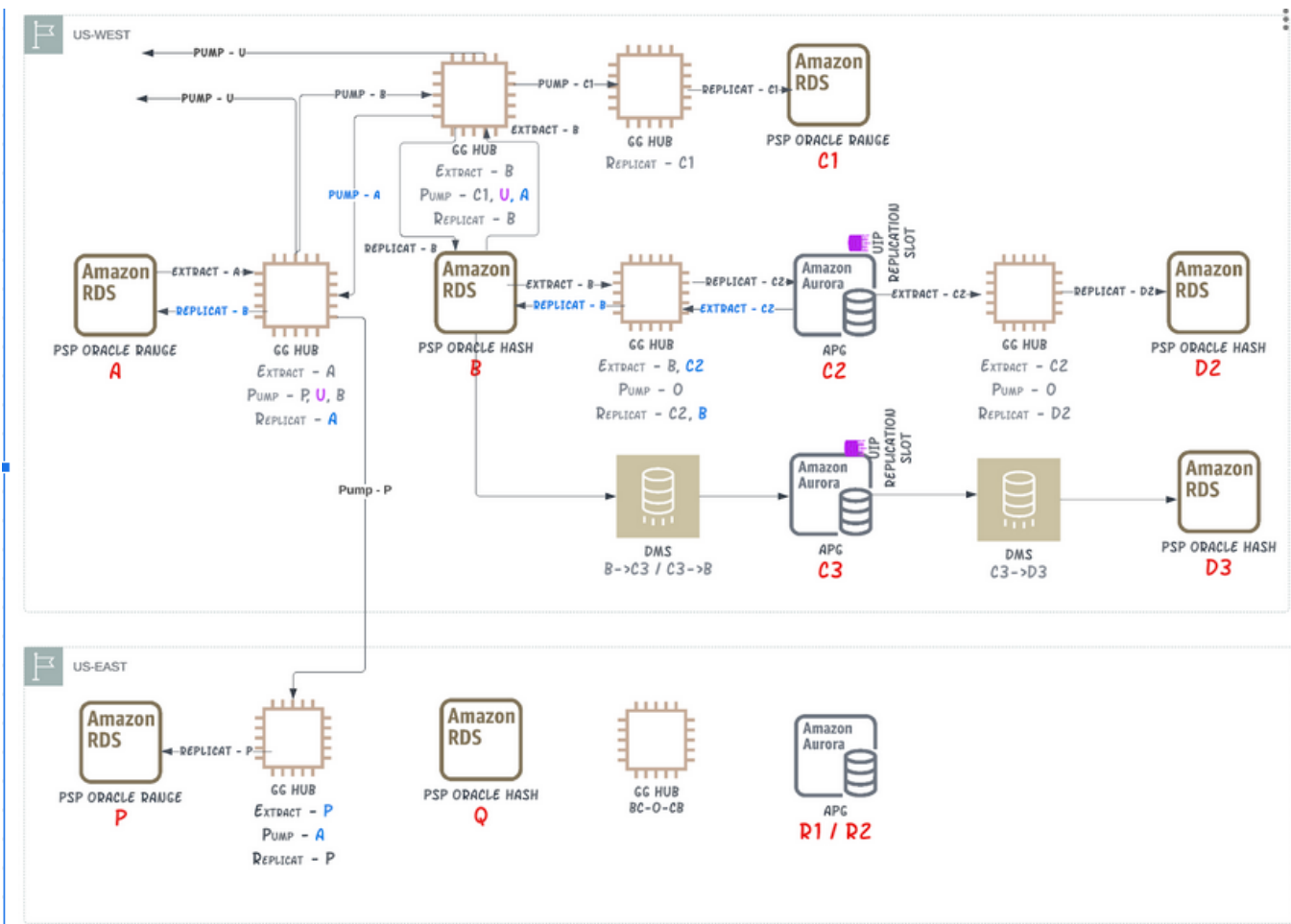


# PSP Prod Monolith Aurora PostgreSQL migration

- Replication Configuration
- Current database object layout
- Assessment
  - Summary
  - Full Report
- Approach
- Schema/Code Changes
  - Detailed Schema Refactor checks
  - Manual Validations Performed
  - Schema Comparison via DB solo or liquibase
- Partitioning Principles
- Query Analysis POC
  - Partitioned Table Join With Non Partitioned Table
  - Partitioned Table Join With Partitioned Table
  - 2 Partitioned Tables Join With Non-Partitioned Table
- Schema Conversion
  - Data Type Mapping
    - NUMBER(19,4) and NUMBER(19,7) Decision
    - NUMBER(10,0) and NUMBER(19,0) Optimization
- Production Replication and Staging Refresh Strategy

## Replication Configuration



| Production and DR |                   |               |      |                   |
|-------------------|-------------------|---------------|------|-------------------|
| West              | PSP Monolith Prod | Instance Name | East | PSP Monolith Prod |

|                   |                               |                        |    |                   |
|-------------------|-------------------------------|------------------------|----|-------------------|
| <b>A</b>          | Oracle (Range)                | pspuwp01               | P  | Oracle (Range)    |
| <b>B</b>          | Oracle (HASH)                 | psphpp02               | Q  | Oracle (HASH)     |
| <b>B(Replica)</b> | Oracle(HASH)                  | psphdg02               | R1 | APG (HASH) GG     |
| <b>C1</b>         | Oracle(Range)                 | psprpp01               | R2 | APG (HASH) DMS    |
| <b>C1'</b>        | Oracle(Range)                 | psprpp02               | S1 | Oracle (HASH) GG  |
| <b>C2</b>         | APG (HASH) GG                 | psp-prod-uw02,pspapg02 | S2 | Oracle (HASH) DMS |
| <b>C3</b>         | APG (HASH) DMS                | psp-prod-mon, psppp01  |    |                   |
| <b>D2</b>         | Oracle (HASH) GG - source C2  | psphpp05               |    |                   |
| <b>D3</b>         | Oracle (HASH) DMS - source C3 | psphpp06               |    |                   |
| <b>D2'</b>        | Oracle (HASH) DMS - source C2 | psphpp07               |    |                   |

## Current database object layout

### Monolith

|    | OWNER  | OBJECT_TYPE        | OBJECT_COUNT |
|----|--------|--------------------|--------------|
| 1  | PSPADM | DATABASE LINK      | 3            |
| 2  | PSPADM | FUNCTION           | 13           |
| 3  | PSPADM | INDEX              | 834          |
| 4  | PSPADM | INDEX PARTITION    | 3758         |
| 5  | PSPADM | INDEX SUBPARTITION | 4830         |
| 6  | PSPADM | LOB                | 30           |
| 7  | PSPADM | LOB PARTITION      | 464          |
| 8  | PSPADM | LOB SUBPARTITION   | 3220         |
| 9  | PSPADM | PACKAGE            | 3            |
| 10 | PSPADM | PACKAGE BODY       | 3            |
| 11 | PSPADM | PROCEDURE          | 28           |
| 12 | PSPADM | SEQUENCE           | 76           |
| 13 | PSPADM | SYNONYM            | 2            |
| 14 | PSPADM | TABLE              | 383          |
| 15 | PSPADM | TABLE PARTITION    | 1931         |
| 16 | PSPADM | TABLE SUBPARTITION | 805          |
| 17 | PSPADM | TRIGGER            | 24           |
| 18 | PSPADM | VIEW               | 5            |

|   | TABLE_NAME                     | No. Of Partitions |
|---|--------------------------------|-------------------|
| 1 | PSP_ENTITY_UPDATE              | 4                 |
| 2 | PSP_ENTRY_DETAIL_RECORD        | 187               |
| 3 | PSP_FINANCIAL_TRANSACTION      | 94                |
| 4 | PSP_FINANCIAL_TRANS_STATE      | 187               |
| 5 | PSP_LEDGER_BALANCE             | 33                |
| 6 | PSP_MONEY_MOVEMENT_TRANSACTION | 94                |
| 7 | PSP_PAYCHECK                   | 32                |

Assessment

Summary

Monolith

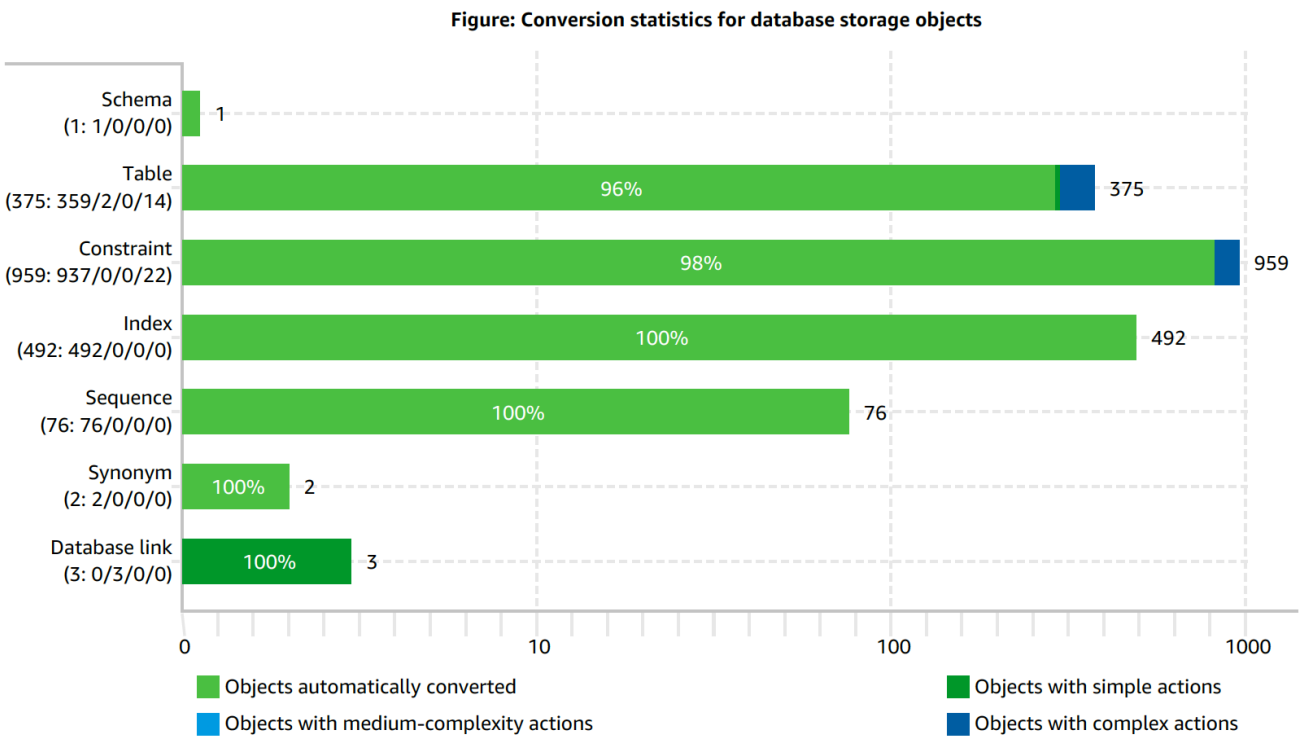
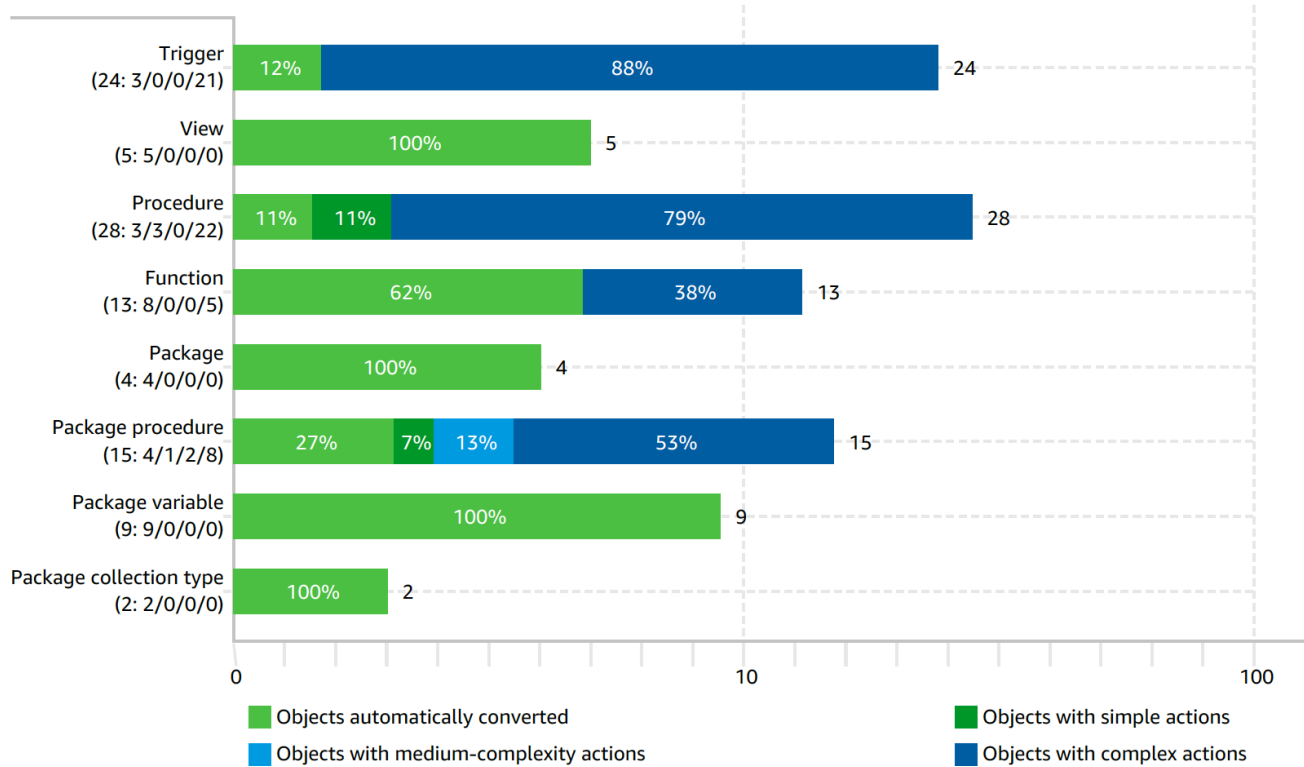


Figure: Conversion statistics for database code objects



No statistics available for database code objects for IBOB/Secondary database.

## Full Report

[psp\\_prod\\_monolith\\_sct\\_migration\\_assessment\\_report\\_aurora\\_postgres.pdf](#)

## Approach

- Migration would be two step approach
  - Schema refactoring on Oracle
  - Migrate to PostgreSQL
- Refactoring includes removal of **realm\_id** column from PK for all tables and adding **company\_fk** as PK to all partitioned tables and its childs.

## Schema/Code Changes

- Primary Key
  - Remove REALM\_ID and Add company\_fk to PK for all partitioned tables.
  - This is to required to have hash partitioning based on company\_fk.
- Partitioning
  - (Why Composite Primary Key?) This is prerequisite to partition by company\_fk in Postgres. [Reference](#)
- Data Integrity
  - If Company\_fk column exists then Populate company\_fk column in all child tables otherwise add company\_fk column in all child tables & populate it.
- Indexes
  - All Primary Key & Foreign Key indexes needs to be changed to reflect Primary Key change.
  - All other Indexes needs to be scrubbed to exclude REALM\_ID.
- Performance
  - Ensure queries come to database with company\_fk filter

|   | Database Objects                 | Change in Oracle Hash                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Change in Postgres Hash                                                                                                                                                                             |
|---|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Table                            | <ul style="list-style-type: none"> <li>Global removal of RELAM_ID reference</li> <li>PK change to composite key</li> <li>Tablespace distribution of tables &amp; Indexes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Global removal of RELAM_ID reference</li> <li>PK change to composite key</li> </ul>                                                                          |
| 2 | Indexes (Secondary & FK indexes) | <ul style="list-style-type: none"> <li>All Partitioned table PK &amp; FK indexes changed to reflect primary key changes</li> <li>Removed REALM_ID references from all indexes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>All Partitioned table PK &amp; FK indexes changed to reflect primary key changes</li> <li>Removed REALM_ID references from all indexes</li> </ul>            |
| 3 | FK Constraints                   | <ul style="list-style-type: none"> <li>All Partitioned table PK &amp; FK constraints changed to reflect primary key changes</li> <li>Removed REALM_ID references from all FK constraints</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>All Partitioned table PK &amp; FK constraints changed to reflect primary key changes</li> <li>Removed REALM_ID references from all FK constraints</li> </ul> |
| 4 | Tables without PK                | <div> <div></div> <div> SCHEMA_NAME, TABLE_NAME<br/> PSPADM, ARCH_PURGE_MAIN_STATS_LIST<br/> PSPADM, AS400_DROPME<br/> PSPADM, BACKUP_SCHEMA_STATS_090911<br/> PSPADM, BACKUP_STATS<br/> PSPADM, BACKUP_STATS_032911<br/> PSPADM, BACKUP_TAB_STATS_090911<br/> PSPADM, CMP3\$100488<br/> PSPADM, CMP3\$15077<br/> PSPADM, CMP4\$15077<br/> PSPADM, DEL_TEST<br/> PSPADM, DISTINCT_COMPANY_FK<br/> PSPADM, EMPTY<br/> PSPADM, PAYCHECK_BACKUP_DATE<br/> PSPADM, PAYROLL_HIST<br/> PSPADM, PET_FEB27<br/> PSPADM, PSP_COMPANY_EVENT_BKP<br/> PSPADM, PSP_COMPANY_NOTE_BKP<br/> PSPADM, PSP_EMPLOYEE_BANK_ACCOUNT_BKP<br/> PSPADM, PSP_FINANCIAL_TRANS_STATE_BKP<br/> PSPADM, PSP_LB_TMP<br/> PSPADM, PSP_LB_TMP_COMP_FK<br/> PSPADM, PSP_LB_TMP_COMP_FK_5000<br/> PSPADM, PSP_LEDGER_BALANCE_BKP<br/> PSPADM, PSP_POSTING_RULE_MAR7<br/> PSPADM, PSP_PROPERTY_AUDIT_BKP<br/> PSPADM, PSP_RPT_PAID_EMPLOYEES<br/> PSPADM, PSP_SAP_METHOD_CALL_BKP<br/> PSPADM, PSP_TRANSACTION_TYPE_MAR7<br/> PSPADM, REP_TEST<br/> PSPADM, STATS_11DEC<br/> PSPADM, TEMP1_PAYCHECK<br/> PSPADM, TEMP_A_PRC_INC<br/> PSPADM, TEMP_BAD_PAYROLL<br/> PSPADM, TEMP_BKP_EDR_JUNE3RD<br/> PSPADM, TEMP_COMPANY_07152010<br/> PSPADM, TEMP_COMPANY_DATAFILE<br/> PSPADM, TEMP_DD4V_PRC_INC<br/> PSPADM, TEMP_DD_PRC_INC<br/> PSPADM, TEMP_ENTITLE_PENDING<br/> PSPADM, TEMP_FIX_NCD<br/> PSPADM, TEMP_FIX_NCD_REACTIVATION<br/> PSPADM, TEMP_FT<br/> PSPADM, TEMP_FUTURE_PAYROLL_DATA<br/> PSPADM, TEMP_GUID_COMPANY<br/> PSPADM, TEMP_MODES_AGENCY_IDS<br/> PSPADM, TEMP_ORSTT_AGENCY_IDS<br/> PSPADM, TEMP_PAYCHECK<br/> PSPADM, TEMP_PEDR_FEB23<br/> PSPADM, TEMP_PFTS_FEB23<br/> PSPADM, TEMP_PFT_FEB23<br/> PSPADM, TEMP_PMT_FEB23<br/> PSPADM, TEMP_PSP_PAYROLL_FRAUD_BATCH </div> </div> |                                                                                                                                                                                                     |

|   |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|---|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|   |                                       | PSPADM, TEMP_RAFFI_SQL_LOG<br>PSPADM, TEMP_SUBSCRIPTIONNUMBER<br>PSPADM, TEMP_TAX_B<br>PSPADM, TEMP_TAX_C<br>PSPADM, TEMP_TAX_CALC_APR12<br>PSPADM, TEMP_TAX_D<br>PSPADM, TEMP_TAX_E<br>PSPADM, TEST2<br>PSPADM, TEST_MMT_ATF<br>PSPADM, TMP_AUTH_USER<br>PSPADM, TMP_AUTH_USER1<br>PSPADM, TMP_PEW_FEB27<br>PSPADM, TMP_PFT_FEB26<br>PSPADM, TMP_PMMT_FEB25<br>PSPADM, TMP_PSP_EDR<br>PSPADM, TMP_PSP_MMT<br>PSPADM, TMP_PTTMD_FEB27<br>PSPADM, TOAD_PLAN_SQL<br>PSPADM, T_BACKFILL_METADATA<br>PSPADM, T_PCEEP_OLD<br>PSPADM, T_PDATL_OLD<br>PSPADM, T_PPPI_OLD<br>PSPADM, T_PPU<br>PSPADM, T_PPU_OLD<br>PSPADM, Z_REDEF_PAYCHECK<br>PSPADM, Z_REDEF_PAYCHECK_SPLIT<br>PSPADM, Z_REDEF_SRC_SYS_TRANSMISSION |  |
| 5 | Tables which are not migrated/skipped | <ul style="list-style-type: none"> <li>PSP_ENTITY_UPDATE_HIST</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
| 6 |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |

## Detailed Schema Refactor checks

- [PSP Schema Refactor Checks](#)

## Manual Validations Performed

- Total object types and its count
- Invalid Objects
- Total Tables
- Partitioned Tables and its partition count
- All Primary Key Columns
- All Primary Key Indexes
- All Index types and its count
- All constraint types and its count'
  - Compare All CHECK and NOT NULL COSNSTRAINTS from production
- Enabled Triggers Count
- Partitioned PK indexes and its partition count
- Local indexes and its partition count
- All FK constraint Names, FK constraint columns and respective Primary table, primary table columns
- Only FK indexes'

## Schema Comparison via DB solo or liquibase

```
--PSP_FINANCIAL_TRANSACTION
create table PSPADM.PSP_FINANCIAL_TRANSACTION
(
    FINANCIAL_TRANSACTION_SEQ VARCHAR2(255) not null
        constraint SYS_C008669
            check ("FINANCIAL_TRANSACTION_SEQ" IS NOT NULL),
    VERSION NUMBER(19) not null
        constraint SYS_C008670
            check ("VERSION" IS NOT NULL),
    CREATOR_ID VARCHAR2(30),
```

```

    CREATED_DATE TIMESTAMP(6) not null
        constraint SYS_C008671
            check ("CREATED_DATE" IS NOT NULL),
    MODIFIER_ID VARCHAR2(30),
    MODIFIED_DATE TIMESTAMP(6) not null
        constraint SYS_C008672
            check ("MODIFIED_DATE" IS NOT NULL),
    REALM_ID NUMBER(19) default -1 not null
        constraint SYS_C008673
            check ("REALM_ID" IS NOT NULL),
    FINANCIAL_TRANSACTION_AMOUNT NUMBER(19,4),
    SETTLEMENT_DATE TIMESTAMP(6),
    SETTLEMENT_TYPE_CD VARCHAR2(255)
        constraint C_PSP_FINANCIAL_TRANSACTION0
            check (SETTLEMENT_TYPE_CD IN
('EFE', 'ACH', 'Wire', 'Cash', 'CheckType', 'Other', 'ApplyForward', 'HPDE', 'TakenOnReturn', 'EFTPS',
'EFTPSDirectDebit', 'EDI')),
    CREDIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
        constraint C_PSP_FINANCIAL_TRANSACTION1
            check (CREDIT_BANK_ACCOUNT_TYPE
IN('Company', 'Employee', 'Intuit', 'TaxAgency', 'Payee')),
    DEBIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
        constraint C_PSP_FINANCIAL_TRANSACTION2
            check (DEBIT_BANK_ACCOUNT_TYPE IN
('Company', 'Employee', 'Intuit', 'TaxAgency', 'Payee')),
    ON_HOLD NUMBER(1),
    SKU VARCHAR2(40),
    SKU_QUANTITY NUMBER(10),
    BILLING_DETAIL_FK VARCHAR2(255),
    CREDIT_BANK_ACCOUNT_FK VARCHAR2(255),
    DEBIT_BANK_ACCOUNT_FK VARCHAR2(255),
    COMPANY_FK VARCHAR2(255),
    PAYROLL_RUN_FK VARCHAR2(255),
    PAYCHECK_SPLIT_FK VARCHAR2(255),
    TRANSACTION_TYPE_FK VARCHAR2(255) not null
        constraint SYS_C008674
            check ("TRANSACTION_TYPE_FK" IS NOT NULL),
    LAW_FK VARCHAR2(255),
    CURRENT_TRANSACTION_STATE_FK VARCHAR2(255),
    ORIGINAL_TRANSACTION_FK VARCHAR2(255),
    MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255),
    ORIGINAL_SETTLEMENT_DATE TIMESTAMP(6),
    REFUND_TYPE VARCHAR2(255 char)
        constraint C_PSP_FINANCIAL_TRANSACTION3
            check (REFUND_TYPE IN
('ApplyForward', 'Immediate', 'TOR')),
    COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
    TAX_PENALTY_INTEREST_FK VARCHAR2(255 char),
    RELATABLE_TRANSACTION_FK VARCHAR2(255 char),
    BILL_PAYMENT_SPLIT_FK VARCHAR2(255 char),
    COMPANY_LAW_FK VARCHAR2(255 char),
    constraint SYS_C008675
        primary key (FINANCIAL_TRANSACTION_SEQ, REALM_ID), -- Remove REALM_ID & add COMPANY_FK. NEW PK
(COMPANY_FK, FINANCIAL_TRANSACTION_SEQ)
        constraint PSP_FINANCIAL_TRANSACTION_FK1
            foreign key (BILLING_DETAIL_FK, REALM_ID) references
PSPADM.PSP_BILLING_DETAIL, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK2
            foreign key (RELATABLE_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION,
-- PK is changing removing REALM_ID and COMPANY_FK to
reflect the same
        constraint PSP_FINANCIAL_TRANSACTION_FK3
            foreign key (ORIGINAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION,
-- PK is changing removing REALM_ID and COMPANY_FK to
reflect the same
        constraint PSP_FINANCIAL_TRANSACTION_FK10
            foreign key (MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION,
-- PK is changing removing REALM_ID and
COMPANY_FK to reflect the same
        constraint PSP_FINANCIAL_TRANSACTION_FK11
            foreign key (COMP_ADJUST_SUBMISSION_FK, REALM_ID)
references PSPADM.PSP_COMP_ADJUST_SUBMISSION, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK13
            foreign key (BILL_PAYMENT_SPLIT_FK, REALM_ID) references
PSPADM.PSP_BILL_PAYMENT_SPLIT, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK14
            foreign key (TAX_PENALTY_INTEREST_FK, REALM_ID) references
PSPADM.PSP_TAX_PENALTY_INTEREST, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK16
            foreign key (COMPANY_LAW_FK, REALM_ID) references PSPADM.
PSP_COMPANY_LAW, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK2
            foreign key (CREDIT_BANK_ACCOUNT_FK, REALM_ID) references
PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
        constraint PSP_FINANCIAL_TRANSACTION_FK3
            foreign key (DEBIT_BANK_ACCOUNT_FK, REALM_ID) references

```

```

PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
constraint PSP_FINANCIAL_TRANSACTION_FK4          foreign key (COMPANY_FK, REALM_ID) references PSPADM.
PSP_COMPANY, -- Remove REALM_ID
constraint PSP_FINANCIAL_TRANSACTION_FK5          foreign key (PAYROLL_RUN_FK, REALM_ID) references PSPADM.
PSP_PAYROLL_RUN, -- Remove REALM_ID
constraint PSP_FINANCIAL_TRANSACTION_FK6          foreign key (PAYCHECK_SPLIT_FK, REALM_ID) references PSPADM.
PSP_PAYCHECK_SPLIT, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
constraint PSP_FINANCIAL_TRANSACTION_FK7          foreign key (TRANSACTION_TYPE_FK, REALM_ID) references
PSPADM.PSP_TRANSACTION_TYPE, -- Remove REALM_ID
constraint PSP_FINANCIAL_TRANSACTION_FK8          foreign key (LAW_FK, REALM_ID) references PSPADM.PSP_LAW,
-- Remove REALM_ID
constraint PSP_FINANCIAL_TRANSACTION_FK9          foreign key (CURRENT_TRANSACTION_STATE_FK, REALM_ID)
references PSPADM.PSP_TRANSACTION_STATE -- Remove REALM_ID )
/
===== Child table =====
--PSP_QBDT_TRANSACTION_INFO
create table PSPADM.PSP_QBDT_TRANSACTION_INFO
(
    QBDT_TRANSACTION_INFO_SEQ VARCHAR2(255 char) not null,
    VERSION_NUMBER(19) not null,
    CREATOR_ID VARCHAR2(30 char),
    CREATED_DATE TIMESTAMP(6) not null,
    MODIFIER_ID VARCHAR2(30 char),
    MODIFIED_DATE TIMESTAMP(6) not null,
    REALM_ID NUMBER(19) default -1 not null,
    AGENCY_NAME VARCHAR2(128 char),
    REFERENCE_NUMBER VARCHAR2(11 char),
    ACCOUNT_NAME VARCHAR2(128 char),
    MEMO VARCHAR2(4000 char),
    ON_SERVICE_NUMBER(1),
    CLEARED VARCHAR2(1 char),
    TRACKING_CLASS VARCHAR2(128 char),
    IS_DELETED NUMBER(1),
    TOKEN_NUMBER(19) default -2,
    LIABILITY_CHECK_FK VARCHAR2(255 char),
    LIABILITY_CHECK_LINE_FK VARCHAR2(255 char),
    MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255 char),
    IS_DIRECT_DEPOSIT NUMBER(1) default 0,
    SYSTEM_GENERATED NUMBER(1) default 0,
    COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
    LIABILITY_ADJUSTMENT_FK VARCHAR2(255 char),
    FINANCIAL_TRANSACTION_FK VARCHAR2(255 char),
    PRIOR_PAYMENT_SUBMISSION_FK VARCHAR2(255 char),
    COMPANY_FK VARCHAR2(255 char),
    QBDT_PAYROLL_TRANSACTION_FK VARCHAR2(255 char),
    QBDT_PAYROLL_TRANS_LINE_FK VARCHAR2(255 char),
    primary key (QBDT_TRANSACTION_INFO_SEQ, REALM_ID), -- remove relam_id from PK
    constraint PSP_QBDT_TRANSACTION_INFO_FK1          foreign key (PRIOR_PAYMENT_SUBMISSION_FK,
REALM_ID) references PSPADM.PSP_PRIOR_PAYMENT_SUBMISSION,
    constraint PSP_QBDT_TRANSACTION_INFO_FK1          foreign key (LIABILITY_CHECK_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_CHECK,
    constraint PSP_QBDT_TRANSACTION_INFO_FK2          foreign key (LIABILITY_CHECK_LINE_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_CHECK_LINE,
    constraint PSP_QBDT_TRANSACTION_INFO_FK3          foreign key (MONEY_MOVEMENT_TRANSACTION_FK,
REALM_ID) references PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK
to reflect the same
    constraint PSP_QBDT_TRANSACTION_INFO_FK4          foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the
same
    constraint PSP_QBDT_TRANSACTION_INFO_FK5          foreign key (COMP_ADJUST_SUBMISSION_FK,
REALM_ID) references PSPADM.PSP_COMP_ADJUST_SUBMISSION,
    constraint PSP_QBDT_TRANSACTION_INFO_FK6          foreign key (LIABILITY_ADJUSTMENT_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_ADJUSTMENT,
    constraint PSP_QBDT_TRANSACTION_INFO_FK7          foreign key (COMPANY_FK, REALM_ID) references
PSPADM.PSP_COMPANY,
    constraint PSP_QBDT_TRANSACTION_INFO_FK8          foreign key (QBDT_PAYROLL_TRANSACTION_FK,
REALM_ID) references PSPADM.PSP_QBDT_PAYROLL_TRANSACTION,
    constraint PSP_QBDT_TRANSACTION_INFO_FK9          foreign key (QBDT_PAYROLL_TRANS_LINE_FK,
REALM_ID) references PSPADM.PSP_QBDT_PAYROLL_TRANS_LINE
)
/

```

```

create index PSPADM.PSP_QBDTTRANSACTIONINFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(PRIOR_PAYMENT_SUBMISSION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK2 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_LINE_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK3 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
MONEY_MOVEMENT_TRANSACTION_FK)
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK4 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(FINANCIAL_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
FINANCIAL_TRANSACTION_FK)
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK5 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(COMP_ADJUST_SUBMISSION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK6 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_ADJUSTMENT_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK7 on PSPADM.PSP_QBDT_TRANSACTION_INFO (COMPANY_FK, REALM_ID); --
remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK8 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANSACTION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK9 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANS_LINE_FK, REALM_ID); --remove REALM_ID

--PSP_TRANSACTION_OFFLOAD_BATCH
create table PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
(
    TRANSACTION_OFFLOAD_BATCH_SEQ VARCHAR2(255 char) not null,
    VERSION_NUMBER(19) not null,
    CREATOR_ID VARCHAR2(30 char),
    CREATED_DATE TIMESTAMP(6) not null,
    MODIFIER_ID VARCHAR2(30 char),
    MODIFIED_DATE TIMESTAMP(6) not null,
    REALM_ID NUMBER(19) default -1 not null,
    OFFLOAD_STATUS_CD VARCHAR2(10 char),
    STATUS_EFFECTIVE_DATE TIMESTAMP(6),
    FINANCIAL_TRANSACTION_FK VARCHAR2(255 char) not null,
    OFFLOAD_BATCH_FK VARCHAR2(255 char) not null,
    primary key (TRANSACTION_OFFLOAD_BATCH_SEQ, REALM_ID), --Remove REALM_ID
constraint PSP_TRANSACTION_OFFLOAD_BA_FK1 foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the
same. Eg (COMPANY_FK, FINANCIAL_TRANSACTION_FK)
    constraint PSP_TRANSACTION_OFFLOAD_BA_FK2 foreign key (OFFLOAD_BATCH_FK, REALM_ID) references
PSPADM.PSP_OFFLOAD_BATCH --remove REALM_ID
)
/

create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK2 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH (OFFLOAD_BATCH_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK1 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
(FINANCIAL_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
FINANCIAL_TRANSACTION_FK)

```

## Partitioning Principles

- Consider table for partitioning whose size is >100G. This is to handle organic & inorganic growth for tables.
- Partition tables by tenant id (company\_fk). This will also help in horizontal scalability of database.
- Consider tables sizes and growth rate (estimated/projected next 3 year and 5 year)
- Keep same number of partitions for logical grouping of tables as per workflows
- Keep individual partition size less than or equal to 50G
- Distribute tables across multiple tablespaces to ensure we do not hit AWS RDS big data file size limit.
- Review partitioned tables for OLTP & OLAP traffic to confirm on number of partitions
- partitioned tables should have company\_fk filter

## Query Analysis POC

- Partitioned Table Join With Non Partitioned Table

| Filter /Join              | Queries                                                                                                                                                                                                                                                                                                                                                                                                | Query Plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Observations                           |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Both +<br>company_fk join | <pre> explain select * from pspadmhashdms. psp_money_movement_transaction on mmt,     pspadmhashdms. psp_atfpayments_to_process atp where mmt. money_movement_transaction_s eq = atp. money_movement_transaction_f k     and mmt.COMPANY_FK ='81ea198c-6fa4-4a19-abcd- 705c93605e43'     and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd- 705c93605e43' ; </pre>                                          | <pre> QUERY PLAN  ----- Merge Join  (cost=1.39..3967.18 rows=2 width=1352)   Merge Cond: ((mtt.money_movement_transaction_seq):: text = (atp.money_movement_transaction_fk)::text)     -&gt;  Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt  (cost=0.70.. 2589.48 rows=2502 width=1153)       Index Cond: ((company_fk)::text = '81ea198c-6fa4- 4a19-abcd-705c93605e43'::text)         -&gt;  Index Scan using psp_atfpayments_to_process_fk1 on psp_atfpayments_to_process atp  (cost=0.69..1368.09 rows=1337 width=199)           Index Cond: ((company_fk)::text = '81ea198c-6fa4- 4a19-abcd-705c93605e43'::text) (6 rows) </pre> | Company_fk<br>join is not<br>required. |
|                           | <pre> explain select * from pspadmhashdms. psp_money_movement_transaction on mmt,     pspadmhashdms. psp_atfpayments_to_process atp where mmt. money_movement_transaction_s eq = atp. money_movement_transaction_f k     and mmt.company_fk = atp. company_fk     and mmt.COMPANY_FK ='81ea198c-6fa4-4a19-abcd- 705c93605e43'     and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd- 705c93605e43' ; </pre> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |

- Partitioned Table Join With Partitioned Table

| Filter /Join | Queries | Query Plan | Observations |
|--------------|---------|------------|--------------|
|--------------|---------|------------|--------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Both +<br>company_fk join | <div data-bbox="233 153 560 667"><pre>select * from pspadmhashdms.   psp_money_movement_transaction mmt,       pspadmhashdms.   psp_financial_transaction ft where mmt.   money_movement_transaction_seq = ft.   money_movement_transaction_fk     and mmt.COMPANY_FK = ft.     COMPANY_FK     and mmt.COMPANY_FK =       '81ea198c-6fa4-4a19-abcd-705c93605e43'     and ft.COMPANY_FK =       '81ea198c-6fa4-4a19-abcd-705c93605e43';</pre></div> <div data-bbox="233 737 560 1199"><pre>select * from pspadmhashdms.   psp_money_movement_transaction mmt,       pspadmhashdms.   psp_financial_transaction ft where mmt.   money_movement_transaction_seq = ft.   money_movement_transaction_fk     and mmt.COMPANY_FK =       '81ea198c-6fa4-4a19-abcd-705c93605e43'     and ft.COMPANY_FK =       '81ea198c-6fa4-4a19-abcd-705c93605e43';</pre></div> | <div data-bbox="589 134 803 159">same plan for both queries</div> <div data-bbox="604 218 664 266">QUERY<br/>PLAN</div> <div data-bbox="604 302 1308 810"><pre>----- ----- ----- Merge Join (cost=1.39..4806.82 rows=22756 width=3289)   Merge Cond: ((ft.money_movement_transaction_fk)::text = (mmt.money_movement_transaction_seq)::text)     -&gt; Index Scan using psp_financial_transaction_fk10_p3 on psp_financial_transaction_p3 ft (cost=0.69..1871.46 rows=1819 width=2136)       Index Cond: ((company_fk)::text = '81ea198c-6fa4- 4a19-abcd-705c93605e43')::text)     -&gt; Materialize (cost=0.70..2595.73 rows=2502 width=1153)       -&gt; Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt (cost=0.70..2589.48 rows=2502 width=1153)         Index Cond: ((company_fk)::text = '81ea198c- 6fa4-4a19-abcd-705c93605e43')::text) (7 rows)</pre></div> | Company_fk<br>join is not<br>required. |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |

• 2 Partitioned Tables Join With Non-Partitioned Table

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 2 Partitioned<br>/Non-partition<br>tables filters<br>/join | <pre> explain select * from pspadmhashdms. psp_money_movement_transa ction mmt,       pspadmhashdms. psp_financial_transaction ft,       pspadmhashdms. psp_atfpayments_to_proces s atp where mmt. money_movement_transactio n_seq = ft. money_movement_transactio n_fk       and mmt. money_movement_transactio n_seq = atp. money_movement_transactio n_fk       and mmt.COMPANY_FK = ft.COMPANY_FK       and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'       and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'       and atp.COMPANY_FK ='81ea198c-6fa4-4a19- abcd-705c93605e43'; </pre> | <pre> QUERY PLAN  ----- ----- ----- Nested Loop  (cost=2.08..3250.17 rows=18 width=3488) -&gt; Merge Join  (cost=1.38..3247.45 rows=1 width=2335)       Merge Cond: ((ft.money_movement_transaction_fk):: text = (atp.money_movement_transaction_fk)::text)       -&gt; Index Scan using psp_financial_transaction_fk10_p3 on psp_financial_transaction_p3 ft  (cost=0.69..1871.46 rows=1819 width=2136)             Index Cond: ((company_fk)::text = '81ea198c- 6fa4-4a19-abcd-705c93605e43'::text)       -&gt; Index Scan using psp_atfpayments_to_process_fk1 on psp_atfpayments_to_process atp  (cost=0.69..1368.09 rows=1337 width=199)             Index Cond: ((company_fk)::text = '81ea198c- 6fa4-4a19-abcd-705c93605e43'::text)       -&gt; Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt  (cost=0.70..2.71 rows=1 width=1153)             Index Cond: (((company_fk)::text = '81ea198c-6fa4- 4a19-abcd-705c93605e43'::text) AND ((money_movement_transaction_seq)::text = (ft. money_movement_transaction_fk)::text)) (9 rows) </pre> | Company<br>_fk join is<br>not<br>required. |
|                                                            | <pre> explain select * from pspadmhashdms. psp_money_movement_transa ction mmt,       pspadmhashdms. psp_financial_transaction ft,       pspadmhashdms. psp_atfpayments_to_proces s atp where mmt. money_movement_transactio n_seq = ft. money_movement_transactio n_fk       and mmt. money_movement_transactio n_seq = atp. money_movement_transactio n_fk       and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'       and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'       and atp.COMPANY_FK ='81ea198c-6fa4-4a19- abcd-705c93605e43'; </pre>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |

# Schema Conversion

## Data Type Mapping

Criteria followed for target datatype w.r.t source NUMER datatype

| Precision(m) | Scale(n) | Oracle      | PostgreSQL       | Comments                           |
|--------------|----------|-------------|------------------|------------------------------------|
| <5           | 0        | NUMBER(m,n) | SMALLINT         | <a href="#">Documentation Page</a> |
| 5 >= m <= 9  | 0        | NUMBER(m,n) | INT              |                                    |
| 10<= m <=18  | 0        | NUMBER(m,n) | BIGINT           |                                    |
| m+n <= 15    | n>0      | NUMBER(m,n) | DOUBLE PRECISION |                                    |
| m+n > 15     | n>0      | NUMBER(m,n) | NUMERIC          |                                    |

### NUMBER(19,4) and NUMBER(19,7) Decision

- Use NUMERIC with same precision and scale as that of Oracle

### NUMBER(10,0) and NUMBER(19,0) Optimization

There is opportunity to optimize numeric(10) and numeric(19) data types further. Based on actual values (Min, Max) data analysis, we are following below criteria for these two

- if source datatype is number(10) and current Max value is more than 100M then use BIGINT else use INTEGER
- if source datatype is number(19) and current Max value is more than 100M. Using NUMERIC(19) else use INTEGER
- Use SMALLINT data type in Postgres for NUMBER(1) in Oracle.
- Use INTEGER data type in Postgres for all VERSION columns in all tables to maintain consistency. This has NUMBER(19) data type in Oracle.
- Using SMALLINT for REALM\_ID column in all tables. We used for Audit db migration as well.

|    | Oracle                      | Postgres                                |
|----|-----------------------------|-----------------------------------------|
| 1  | CLOB                        | TEXT                                    |
| 2  | NUMBER(1)                   | SMALLINT                                |
| 3  | NUMBER(10)                  | BIGINT                                  |
| 4  | NUMBER(19)                  | NUMERIC(19,0)                           |
| 5  | NUMBER(19) NOT NULL         | NUMERIC(19,0) NOT NULL                  |
| 6  | NUMBER(19,7)                | NUMERIC(19,7)                           |
| 7  | TIMESTAMP(6)                | timestamp(6) without time zone          |
| 8  | TIMESTAMP(6) NOT NULL       | timestamp(6) without time zone NOT NULL |
| 9  | VARCHAR2(100 Char)          | character varying(100)                  |
| 10 | VARCHAR2(20 Char)           | character varying(20)                   |
| 11 | VARCHAR2(200 Char)          | character varying(200)                  |
| 12 | VARCHAR2(255 Char)          | character varying(255)                  |
| 13 | VARCHAR2(255 Char) NOT NULL | character varying(255) NOT NULL         |
| 14 | VARCHAR2(256 Char)          | character varying(256)                  |
| 15 | VARCHAR2(30 Char)           | character varying(30)                   |
| 16 | VARCHAR2(40 Char)           | character varying(40)                   |
| 17 | VARCHAR2(400 Char)          | character varying(400)                  |
| 18 | VARCHAR2(4000 Char)         | character varying(4000)                 |

## Production Replication and Staging Refresh Strategy

